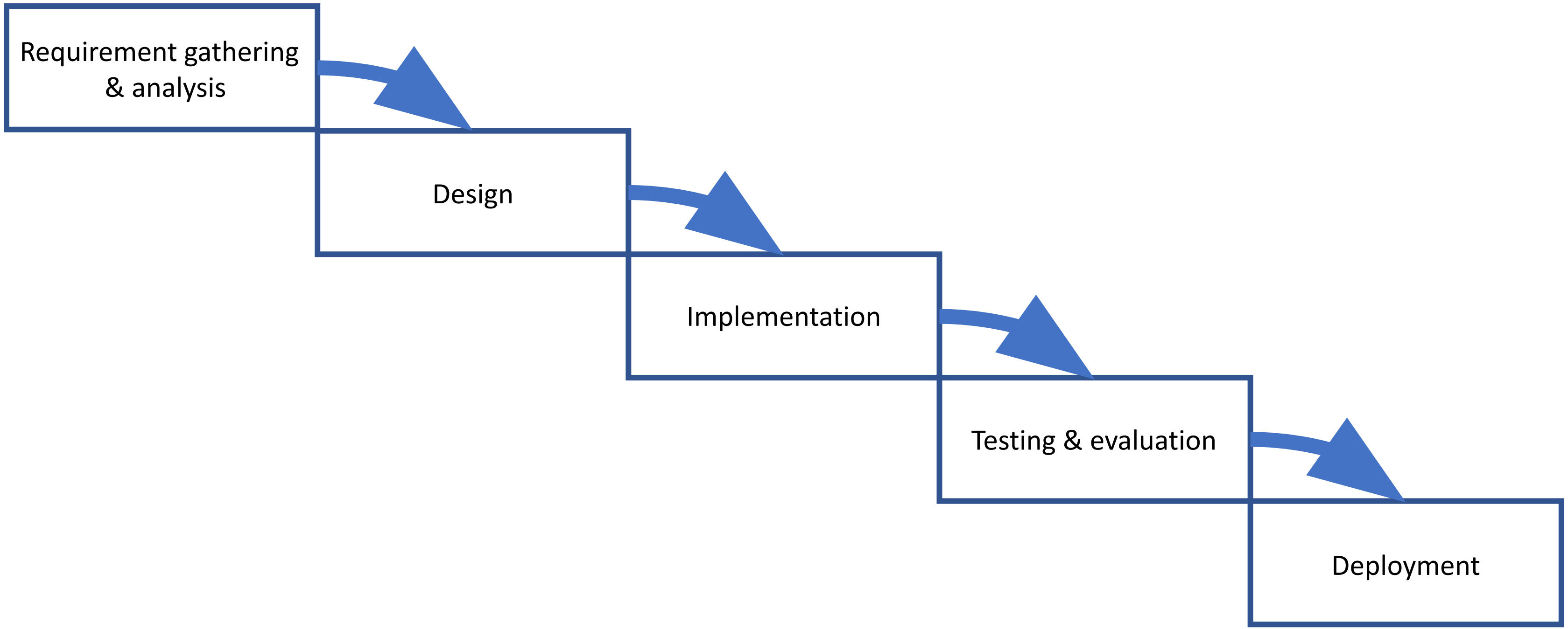
CS 3430 Database

ER Model

Tinghao Feng

Remember: Waterfall Process Model



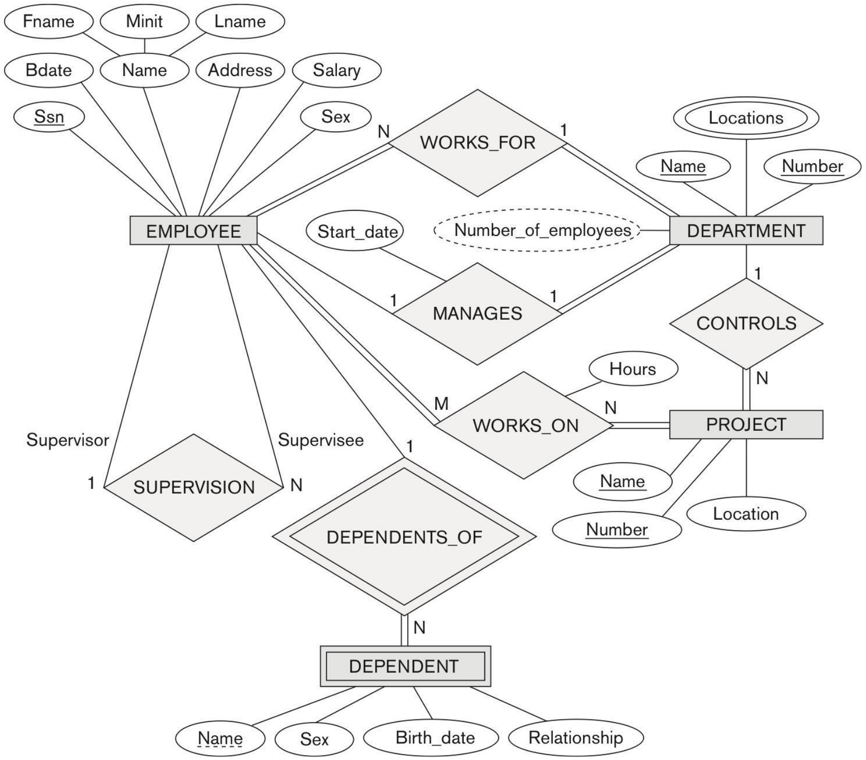
Design

- Requirements → conceptual design/schema
  - Identify things/entities (nouns)
  - Identify relationships among things (verbs)
  - Identify attributes (nouns, adjectives)
- Conceptual schema → logical design/schema
  - Map conceptual model to implementation model
- Logical design schema → physical design/schema
  - Choose suitable DBMS
  - Write constructs in chosen DBMS (DDL)

ER (Entity-Relationship) Model

A popular high-level conceptual data model

COMPANY ER Schema Diagram



Entity And Attributes

Entity

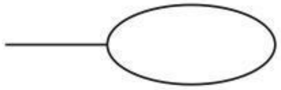
- An entity is something of importance to a user or organization that needs to be represented in a database
- An entity represents one theme, topic, or business concept
- In the ER model, entities are restricted to things that can be represented using a single table

- Notation: 

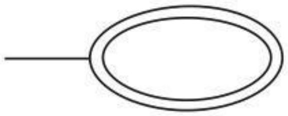
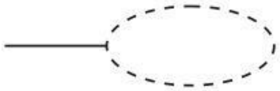
Entity Type And Entity Set

- Entity type
  - Entities with the same basic attributes are grouped or typed into an entity type
- Entity set (collection)
  - A collection of entities of an entity type
  - The entity set is the current *state* of the entities of that type that are stored in the database

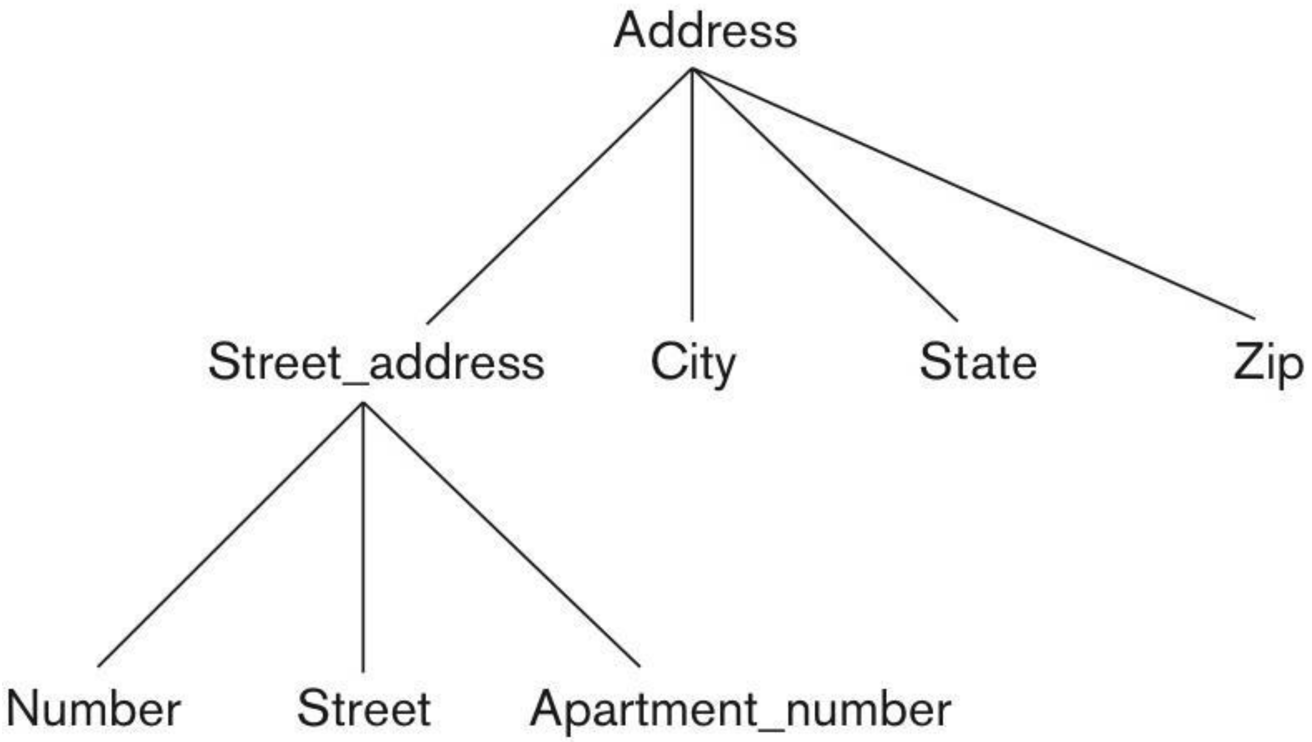
Attribute

- Attributes are properties used to describe an entity
- Each attribute has a value set (or data type) associated with it
- Notation: 

Type Of Attributes

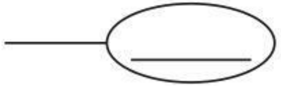
- Simple
- Composite
- Multi-valued: 
- Derived: 

Example Of A Composite Attribute



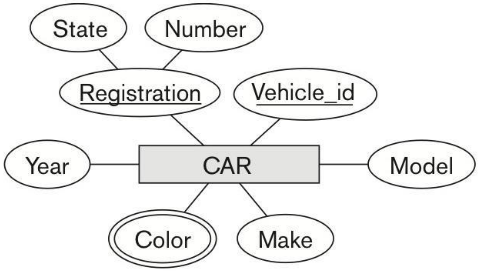
Key Attributes

- An attribute of an entity type for which each entity must have a unique value is called a ***key attribute*** of the entity type
- A key attribute may be composite
- An entity type may have more than one key

- Notation: 

Entity Type CAR With Two Keys

(a)



(b)

CAR
Registration (Number, State), Vehicle_id, Make, Model, Year, {Color}

CAR$_1$
((ABC 123, TEXAS), TK629, Ford Mustang, convertible, 2004 {red, black})

CAR$_2$
((ABC 123, NEW YORK), WP9872, Nissan Maxima, 4-door, 2005, {blue})

CAR$_3$
((VSY 720, TEXAS), TD729, Chrysler LeBaron, 4-door, 2002, {white, blue})

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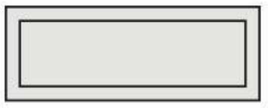
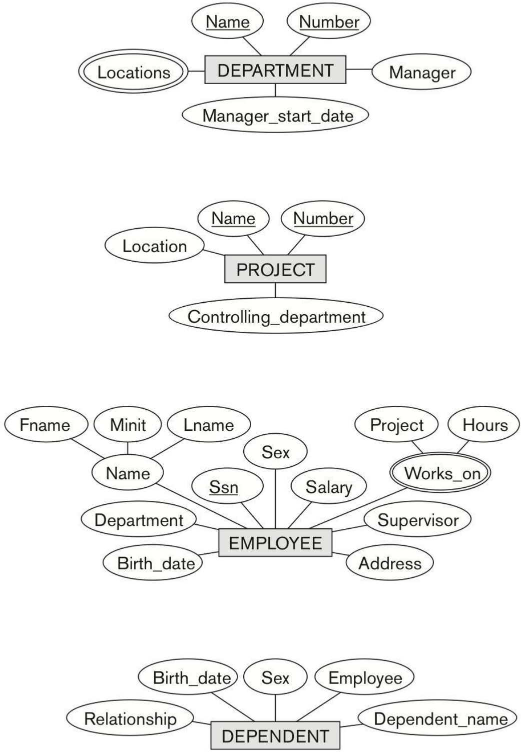
Weak Entity Types

- An entity that does not have a key attribute and that is identification-dependent on another entity type.

- Notation of weak entity type: 

Initial Design Of Entity Types

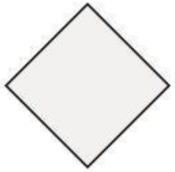


Main Concepts Of ER Model

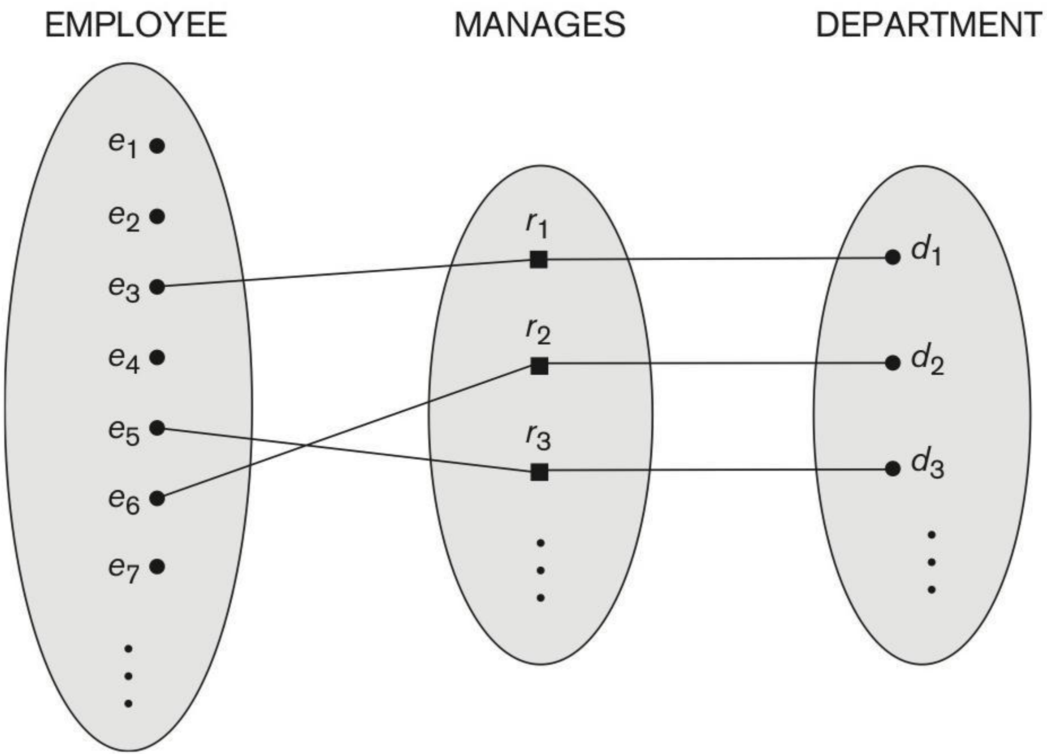
- The initial design is typically not complete
- ER model has three main concepts:
  - Entity
  - Attribute
  - Relationship
- Some aspects in the requirements will be represented as relationships

Relationship

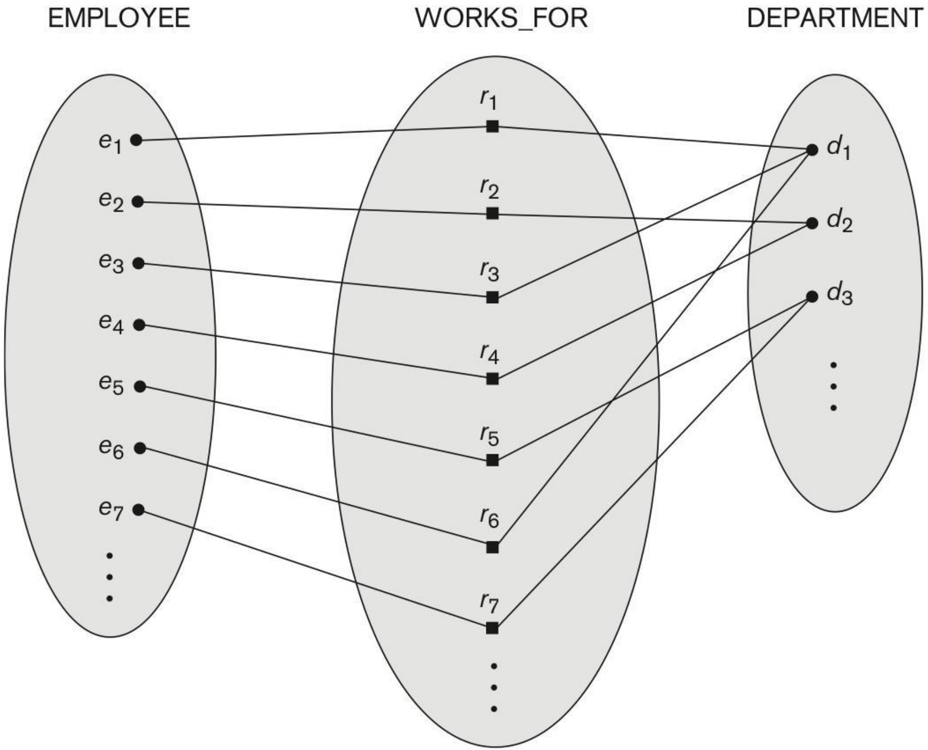
Relationship Type And Relationship Set

- A **relationship** relates two or more distinct entities with a specific meaning.
- Relationships of the same type are grouped or typed into a **relationship type**.
- The degree of a relationship type is the number of participating entity types.
- **Relationship set** is the current state of a relationship type.
- Notation: 

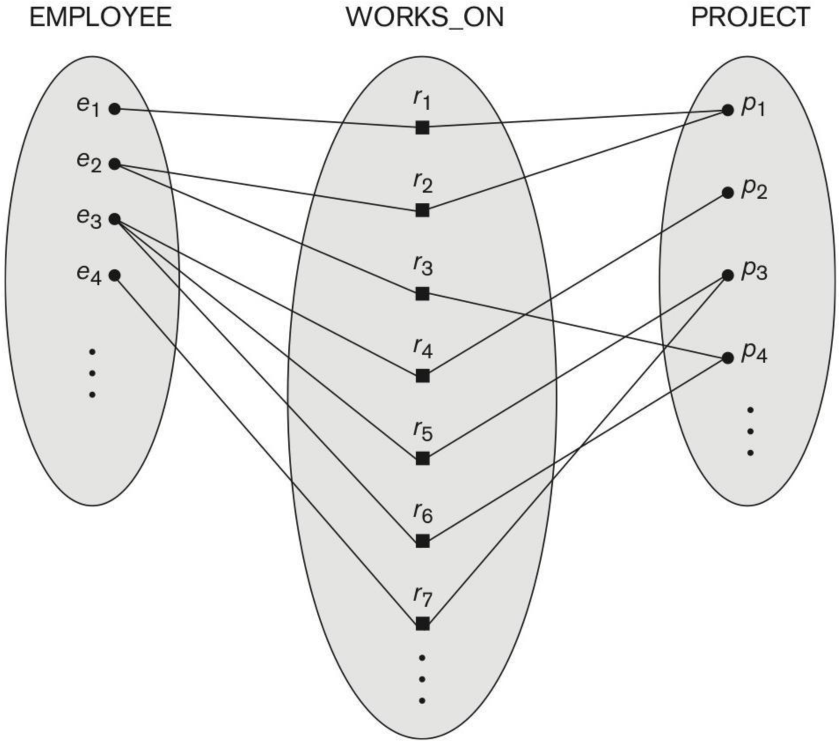
One-to-one (1:1) Relationship



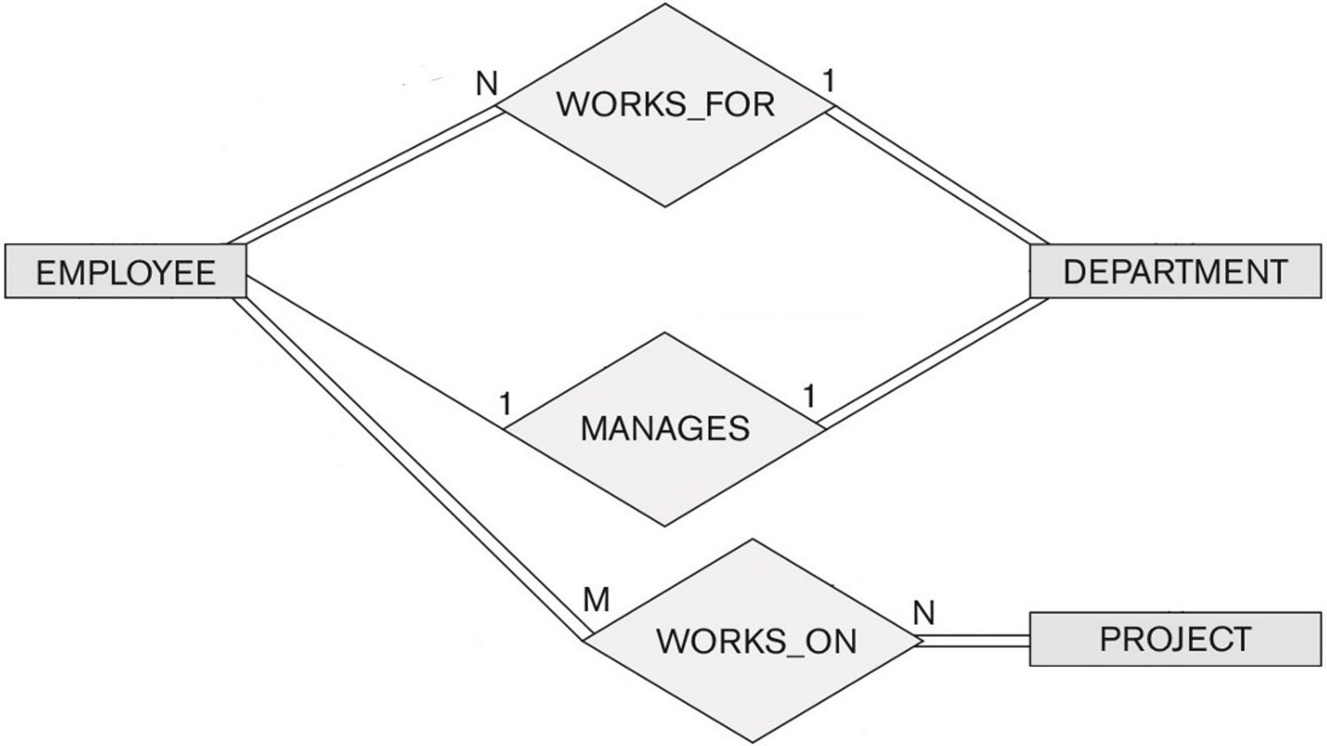
Many-to-one (N:1) Relationship



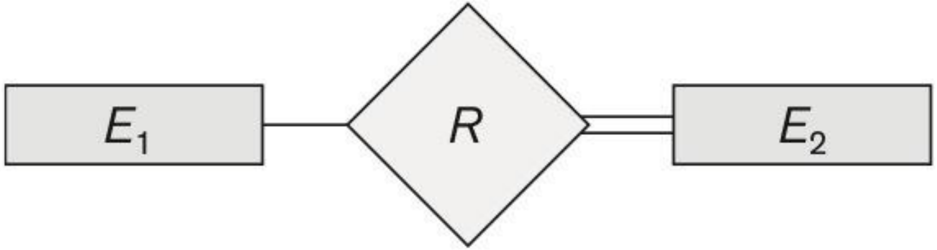
Many-to-many (M:N) Relationship



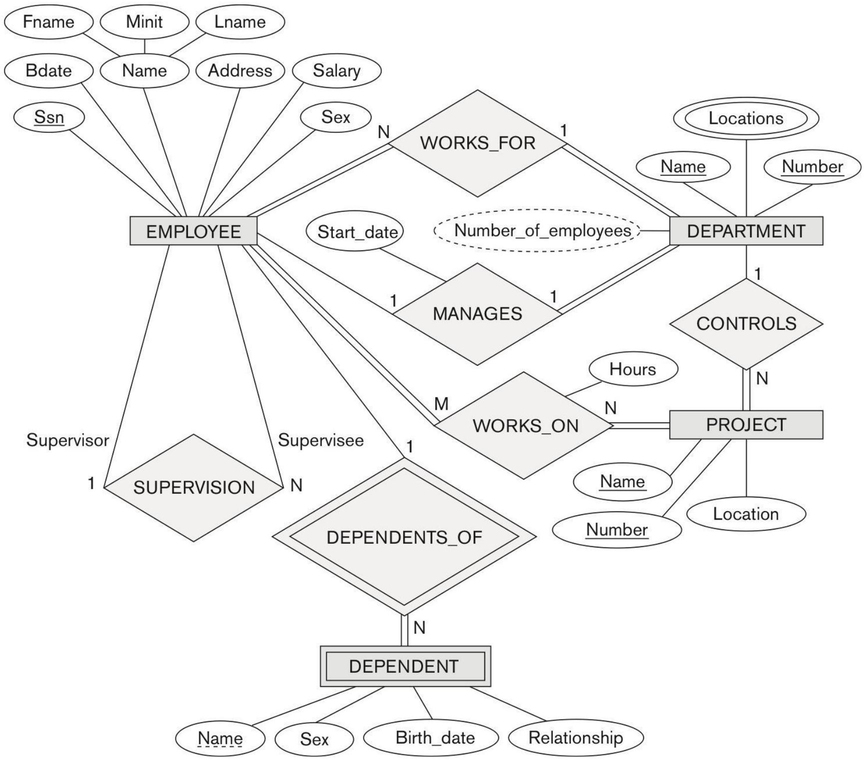
Cardinality Ratio Constraints (specifies maximum participation)



Participation Constraint



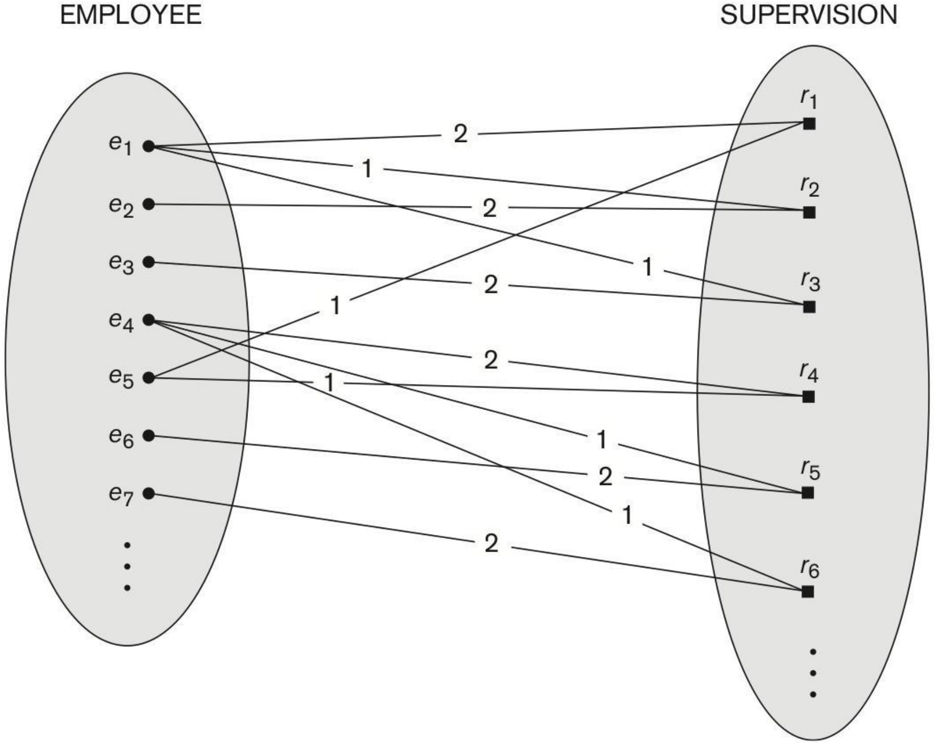
How Many Relationships Can You Find?



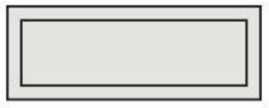
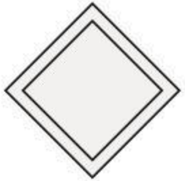
Attributes Of Relationship Types

- A relationship type can have attributes
- Most relationship attributes are used with M:N relationships

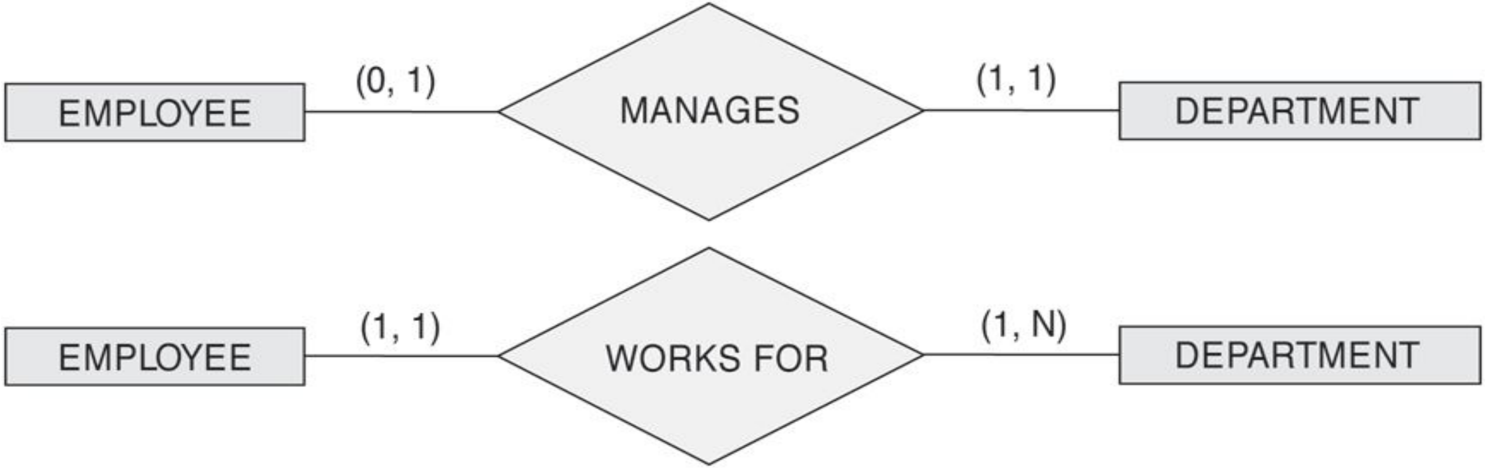
Recursive Relationship Type (Self-referencing)



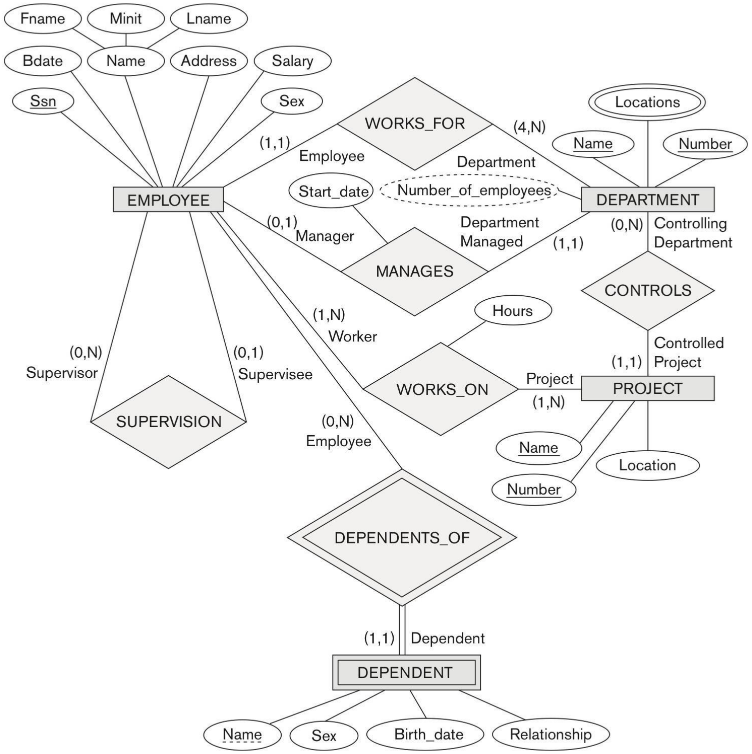
Weak Entity Types

- An entity that does not have a key attribute and that is identification-dependent on another entity type.
- A weak entity must participate in an identifying relationship type with an owner or identifying entity type
- Notation of weak entity type: 
- Notation of identifying relationship: 

(min,max) Notation For Relationship Constraints



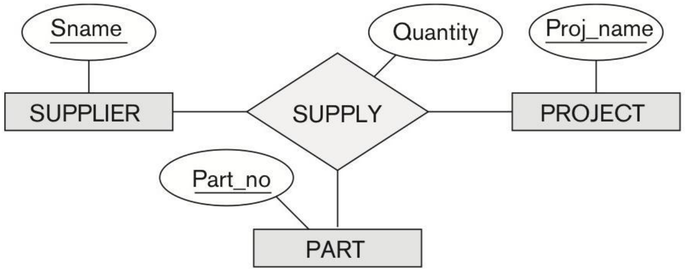
(min, max) Notation For The Example

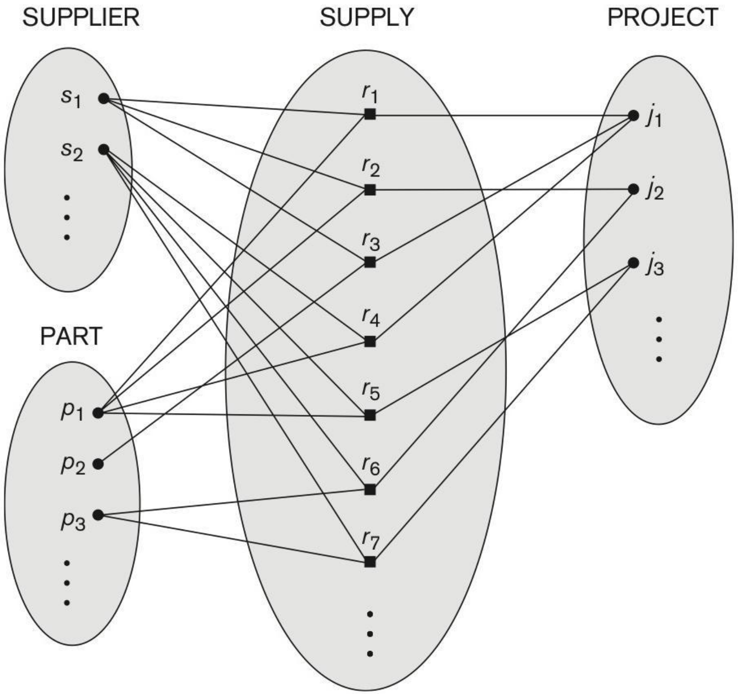


Relationships of Higher Degree

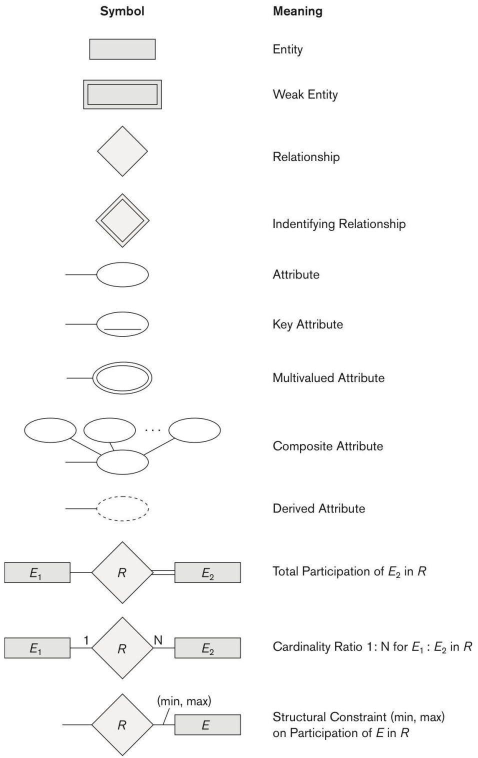
- Relationship types of degree 2 are called binary
- Relationship types of degree 3 are called ternary
- Relationship types of degree n are called n-ary
- Constraints are harder to specify for higher-degree relationships (n > 2) than for binary relationships

Example of a ternary relationship

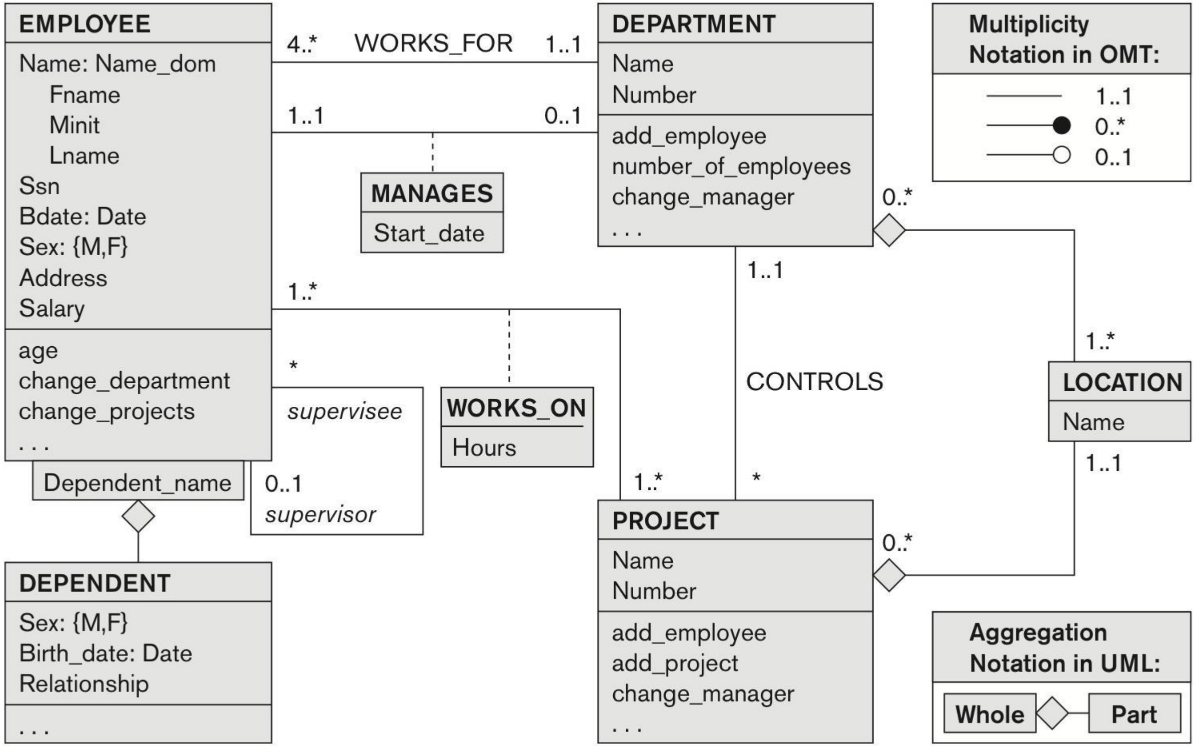




Summary Of Notation For ER Diagrams



UML Class Diagram For The Example



University Database Conceptual Model